Small Image Recognition
Neural Networks on MNIST

MLP vs. CNN vs. Transformer Encoder

github.com/byy9512/ai-hw-summer-2026-nn

AI Homework — Summer 2026

MNIST digit classification

Train only on the train split. Test only on the test split.

Dataset	MNIST — torchvision.datasets.MNIST
Images	28×28 grayscale, 10 classes (digits 0–9)
Split	60,000 train images / 10,000 test images
Normalization	mean = 0.1307, std = 0.3081

ARCHITECTURE

Three models, one task

Each takes a (B, 1, 28, 28) image batch and outputs (B, 10) class logits

MLP	Flatten -> 2 fully-connected layers	`102K params`
CNN	2 convolution + pooling blocks	`422K params`
Transformer	Image patches + self-attention	`139K params`

Each model also has a revised variant:

MLP + BatchNorm & an extra layer · CNN + BatchNorm · Transformer mean-pools patches instead of a [CLS] token

Key difference: MLP has no spatial awareness, CNN builds it in architecturally,
the Transformer must learn it from data via self-attention.

Train & test

Adam optimizer, cross-entropy loss — trained on train split, evaluated on test split only

Optional augmentation (train split only):

random rotation + random translate/scale + random erasing — test split always stays clean

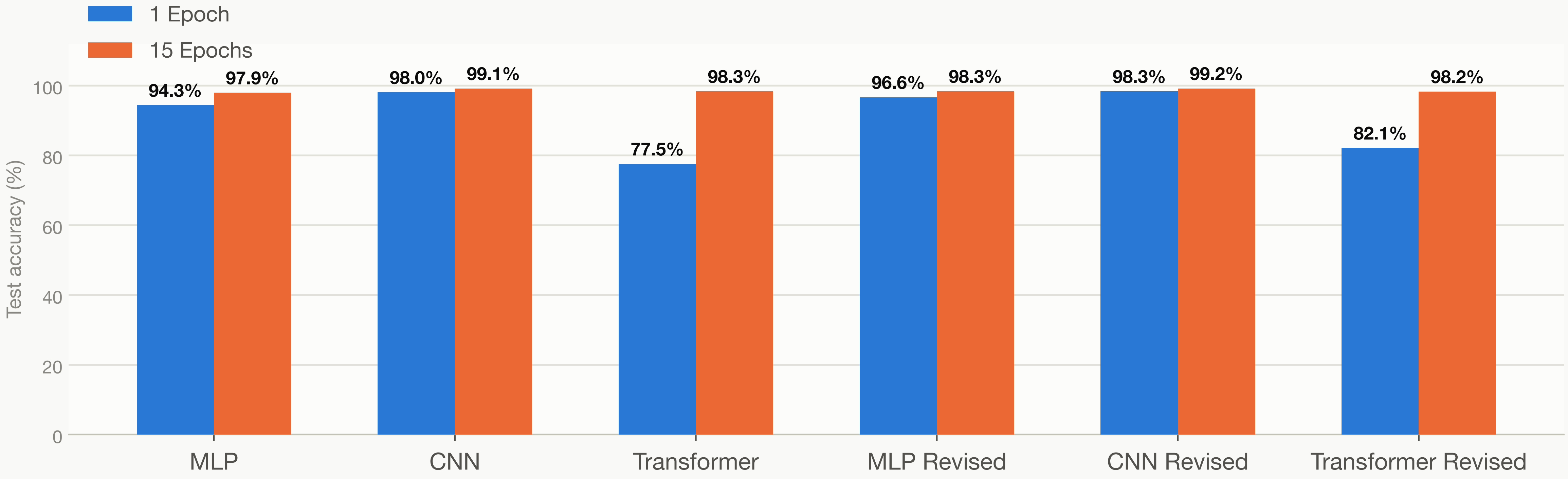
```
python3 train.py cnn --epochs 15 --batch-size 128
```

```
python3 test.py cnn
```

RESULTS

Training budget changes the ranking

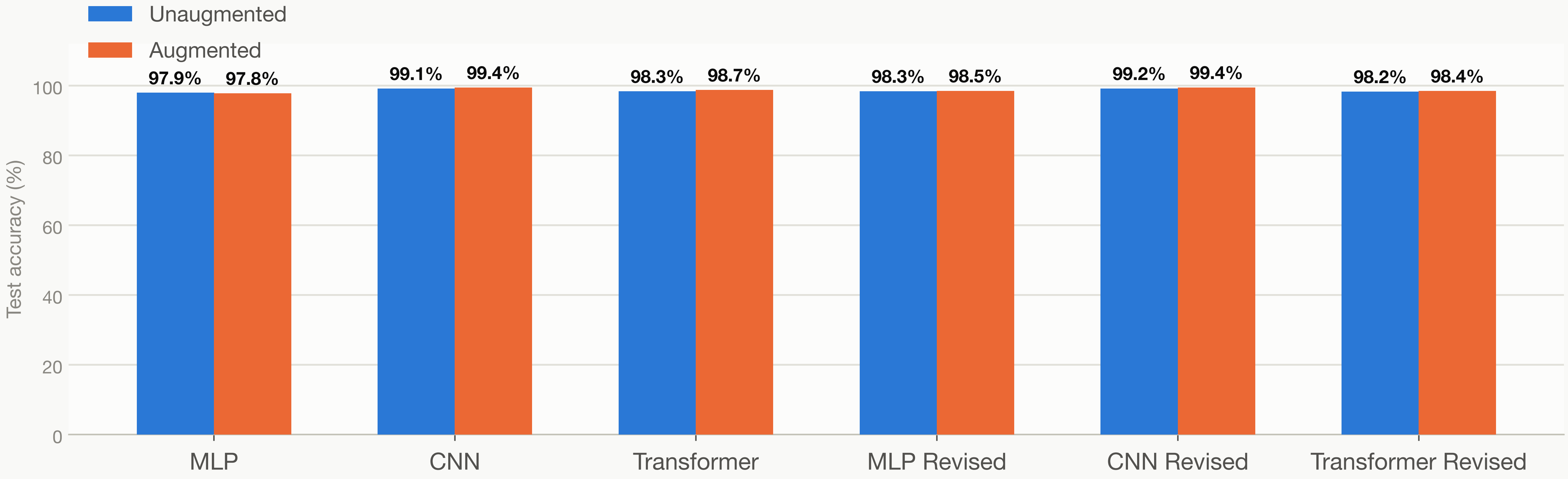
1 epoch vs. 15 epochs, same models, same data



Transformer gains +20.8 pts (77.5% -> 98.3%) — its early weakness was a lack of training time, not a worse architecture. CNN was already near its ceiling at epoch 1.

Augmentation pays off — with enough training

15 epochs, unaugmented vs. augmented (at 1 epoch, augmentation hurt every model)



Best model overall: CNN Revised + Augmentation — 99.44% test accuracy.

Augmentation now helps 5 of 6 models; only plain MLP is flat.

Takeaways

Training budget matters more than architecture choice.

Given enough epochs, the Transformer closes almost its entire gap with the CNN.

Augmentation is a longer-horizon investment.

It hurt every model at 1 epoch, but helped 5 of 6 at 15 epochs.

Best result: CNN Revised + Augmentation — 99.44%.

Strong inductive bias, plus BatchNorm, plus augmentation — stacked improvements win.

Thank you

Code, training/testing scripts, and full results write-up:

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